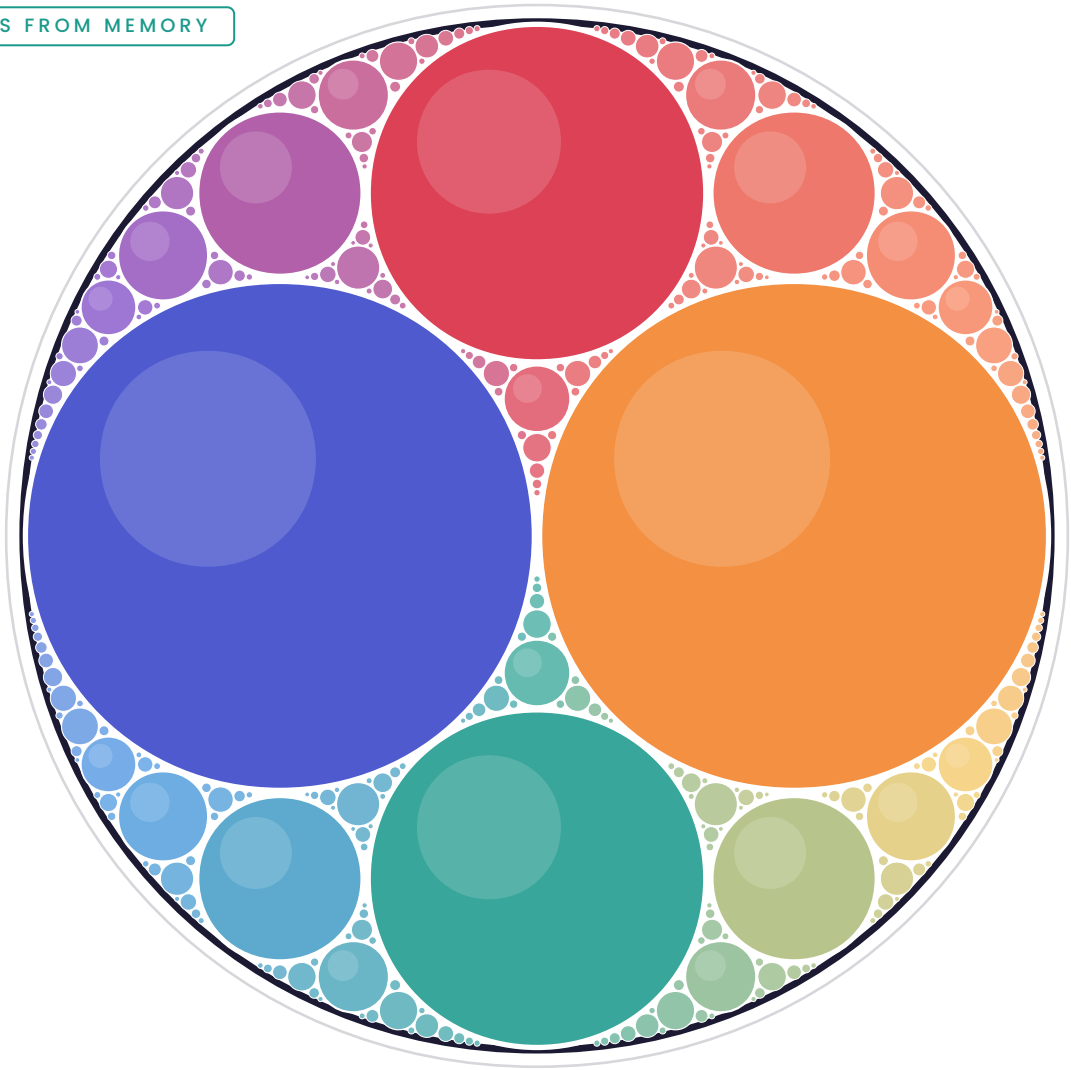
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS FROM MEMORY



NAME _____

CLASS _____

APOLLONIAN GASKET
each circle fills a gap between three touching circles; every curvature (1 ÷ radius) is an integer

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Between which two whole numbers does this square root lie?

$$\sqrt{37}$$

METHOD Write the rule: **estimating square roots**

FROM MEMORY

WORKING

ANSWER

2 Solve.

$$6k + 6 = 3k + 27$$

METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

3 Evaluate.

$$-4 - (-2)$$

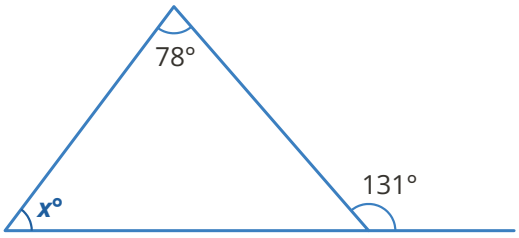
METHOD Write the rule: **subtracting a negative**

FROM MEMORY

WORKING

ANSWER

4 Find the value of x.



FORMULA Write the formula: **exterior angle of a triangle**

FROM MEMORY

SUBSTITUTE

ANSWER

5 20% of the students at a school walk to school. That is 12 students. How many students are at the school?

METHOD Write the rule: **the unitary method**

FROM MEMORY

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The price of a pair of sneakers is $200 before GST. How much is the GST (10%)?

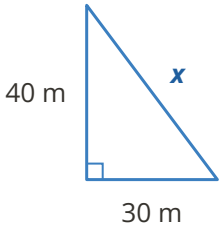
FORMULA Write the formula: **GST**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Find the length of the hypotenuse, *x*.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Find the range of these goal tallies.

Goals	9	8	18	3	17

FORMULA Write the formula: **the range**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Solve.

$$y^2 + 7 = 16$$

METHOD Write the rule: **equations of the form $ax^2 = c$**

FROM MEMORY

WORKING

ANSWER

5 Write 40 as a product of prime factors in index form.

METHOD Write the rule: **prime factors in index form**

FROM MEMORY

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Priya says $9^6 \times 9^5 = 81^{11}$. Explain the mistake. EXPLAIN

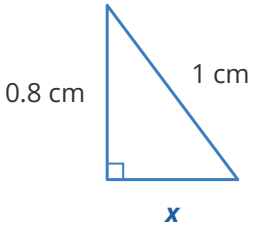
FORMULA Write the formula: **index law for multiplying**

FROM MEMORY

WORKING

EXPLAIN

2 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Write an algebraic expression for:

m divided by 4

METHOD Write the rule: **writing algebraic expressions**

FROM MEMORY

WORKING

ANSWER

4 The price of a surf lesson changed from \$80 to \$112. Find the percentage increase.

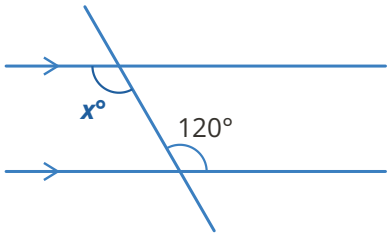
FORMULA Write the formula: **percentage change**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Write the formula: **alternate angles**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 A triangle has sides 5 cm, 12 cm and 13 cm. Is it right-angled?

FORMULA Write the formula: **converse of Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$3^6 \div 3^3$$

FORMULA Write the formula: **index law for dividing**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Which has the larger area: a 9 m by 2 m rectangle or a square with sides of 4 m? Explain. EXPLAIN

FORMULA Write the formula: **area of a rectangle and a square**

FROM MEMORY

WORKING

EXPLAIN

4 Solve. Give your answer as a fraction or a decimal.

$$4a + 1 = 12$$

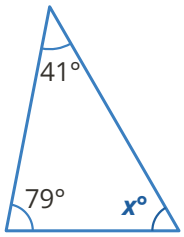
METHOD Write the rule: **solving equations**

FROM MEMORY

WORKING

ANSWER

5 Find the value of x.



FORMULA Write the formula: **angle sum of a triangle**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$4x + 1 = 3x + 10$$

METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

2 Factorise.

$$12x - 28$$

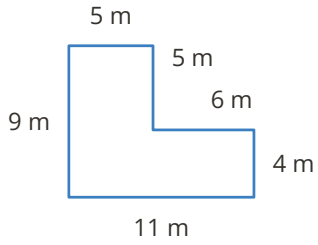
METHOD Write the rule: **factorising**

FROM MEMORY

WORKING

ANSWER

3 Find the area of this shape.



METHOD Write the rule: **area of a composite shape**

FROM MEMORY

WORKING

ANSWER

4 Simplify. Write your answer as a single power.

$$(3^2)^4$$

FORMULA Write the formula: **power of a power**

FROM MEMORY

SUBSTITUTE

ANSWER

5 A boogie board costs $50. It is on sale at 50% off. Find the sale price.

METHOD Write the rule: **discounts**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

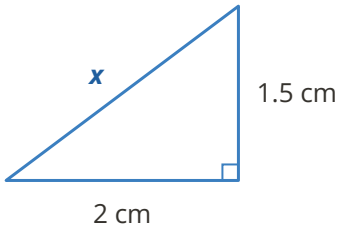
TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 Find the length of the hypotenuse, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Decrease 300 L by 10%.

METHOD Write the rule: **percentage decrease**

FROM MEMORY

WORKING

ANSWER

3 Calculate. Give your answer in simplest form.

$$\frac{6}{10} - \frac{3}{10}$$

METHOD Write the rule: **adding, subtracting and multiplying fractions**

FROM MEMORY

WORKING

ANSWER

4 Complete the table for $y = 3x + 5$.

x	1	2	3	4
y		11	14	

METHOD Write the rule: **tables of values**

FROM MEMORY

WORKING

ANSWER

5 Simplify. Write your answer as a single power.

$$4^6 \times 4^2$$

FORMULA Write the formula: **index law for multiplying**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$-5(4p - 8)$$

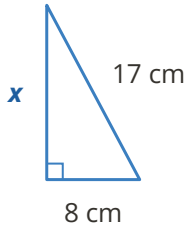
FORMULA Write the formula: **expanding brackets**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Write 20 as a product of its prime factors.

METHOD Write the rule: **prime factors (factor tree)**

FROM MEMORY

WORKING

ANSWER

4 Evaluate.

$$3^2 + 8^0$$

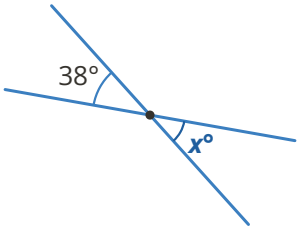
FORMULA Write the formula: **the zero index**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Write the formula: **vertically opposite angles**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The canteen buys a box of 10 drinks for $20 and sells it for $22. Find the profit as a percentage of the cost price.

FORMULA Write the formula: **profit and loss as a percentage**

FROM MEMORY

SUBSTITUTE

ANSWER

2 The angle sum of a polygon with n sides is $S°$. Find n when $S = 720$.

FORMULA $S = 180(n - 2)$

COPY

SUBSTITUTE

ANSWER

3 For these scores, which is larger: the mean or the median? Explain. EXPLAIN

Scores	2	25	6	5	7

FORMULA Write the formula: **the mean**

FROM MEMORY

WORKING

EXPLAIN

4 Simplify. Write your answer as a single power.

$$9^4 \div 9^2$$

FORMULA Write the formula: **index law for dividing**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Expand and simplify.

$$5(n + 6) + 3$$

METHOD Write the rule: **expanding and simplifying**

FROM MEMORY

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

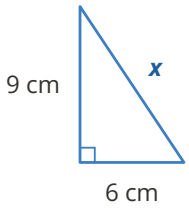
TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the hypotenuse, x, correct to 1 decimal place.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Solve.

$$4x^2 = 16$$

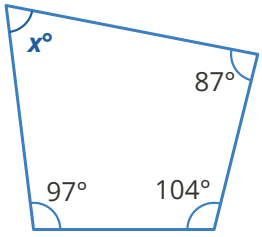
METHOD Write the rule: **equations of the form $ax^2 = c$**

FROM MEMORY

WORKING

ANSWER

3 Find the value of x.



FORMULA Write the formula: **angle sum of a quadrilateral**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Evaluate.

$$9^2$$

FORMULA Write the formula: **index notation**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Increase 300 kg by 20%.

METHOD Write the rule: **percentage increase**

FROM MEMORY

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$4a^2 + 7a - 2a^2 + 4a$$

METHOD Write the rule: **like terms**

FROM MEMORY

WORKING

ANSWER

2 20% of a number is 8. What is the number?

METHOD Write the rule: **the unitary method**

FROM MEMORY

WORKING

ANSWER

3 Simplify.

$$9a + 4b - 4a$$

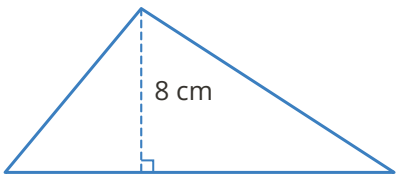
METHOD Write the rule: **simplifying like terms**

FROM MEMORY

WORKING

ANSWER

4 Find the area of this triangle.



FORMULA Write the formula: **area of a triangle**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Solve.

$$5n + 5 = 3n + 23$$

METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the value of this expression when $a = -4$ and $b = 4$.

$$4(a + b)$$

METHOD Write the rule: **substitution**

FROM MEMORY

WORKING

ANSWER

2 Grace thinks of a number, multiplies it by 3 and adds 8. The answer is 32. Write and solve an equation to find the number.

METHOD Write the rule: **equation problems**

FROM MEMORY

WORKING

ANSWER

3 Simplify. Write your answer as a single power.

$$(4^3)^2$$

FORMULA Write the formula: **power of a power**

FROM MEMORY

SUBSTITUTE

ANSWER

4 A letter is chosen at random from this word. Find P(N).

D O L P H I N

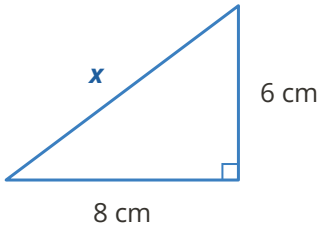
FORMULA Write the formula: **probability of an event**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Find the length of the hypotenuse, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Factorise.

$$36m - 45$$

METHOD Write the rule: **factorising**

FROM MEMORY

WORKING

ANSWER

2 A triangle has sides 8 m, 9 m and 12 m. Is it right-angled?

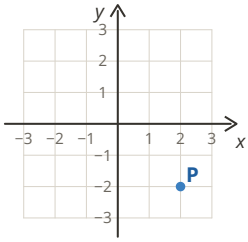
FORMULA Write the formula: **converse of Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Write the coordinates of the point P.



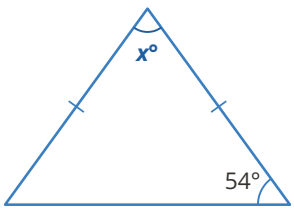
METHOD Write the rule: **coordinates**

FROM MEMORY

WORKING

ANSWER

4 Find the value of x. Give a reason. REASON



FORMULA Write the formula: **isosceles triangle**

FROM MEMORY

SUBSTITUTE

REASON

ANSWER

5 The price of a school bag is $90 before GST. Add 10% GST to find the price including GST.

FORMULA Write the formula: **GST**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Priya says $4^2 \times 4^4 = 16^6$. Explain the mistake. EXPLAIN

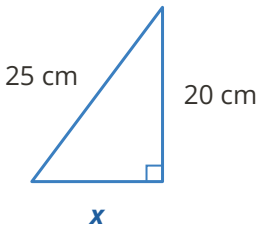
FORMULA Write the formula: **index law for multiplying**

FROM MEMORY

WORKING

EXPLAIN

2 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Solve.

$$5k + 10 = 2k + 13$$

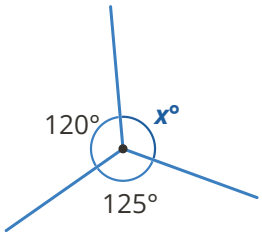
METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

4 Find the value of x.



FORMULA Write the formula: **angles at a point**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Find the median of these netball scores.

Points	18	14	7	16	3

FORMULA Write the formula: **the median**

FROM MEMORY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$30x^2 \div 6x$$

METHOD Write the rule: **multiplying and dividing terms**

FROM MEMORY

WORKING

ANSWER

2 The price of a surf lesson changed from $60 to $84. Find the percentage increase.

FORMULA Write the formula: **percentage change**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Write 48 as a product of prime factors in index form.

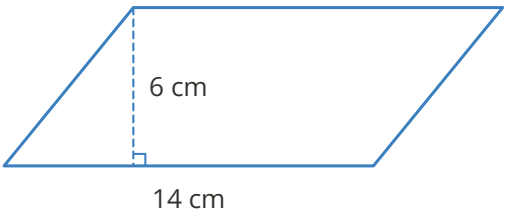
METHOD Write the rule: **prime factors in index form**

FROM MEMORY

WORKING

ANSWER

4 Find the area of this parallelogram.



FORMULA Write the formula: **area of a parallelogram**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Use the formula to find w when $P = 30$ and $l = 10$.

FORMULA $P = 2l + 2w$

COPY

SUBSTITUTE

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$3(p - 1)$$

FORMULA Write the formula: **expanding brackets**

FROM MEMORY

SUBSTITUTE

ANSWER

2 A wetsuit costs $390. It is on sale at 40% off. Find the sale price.

METHOD Write the rule: **discounts**

FROM MEMORY

WORKING

ANSWER

3 Calculate.

$$\frac{3}{5} \text{ of } \$35$$

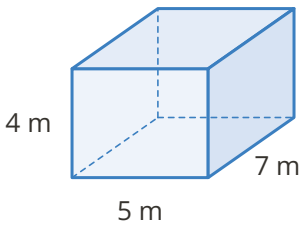
FORMULA Write the formula: **fraction of a quantity**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Find the volume of this rectangular prism.



FORMULA Write the formula: **volume of a rectangular prism**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Simplify. Write your answer as a single power.

$$7^3 \div 7^2$$

FORMULA Write the formula: **index law for dividing**

FROM MEMORY

SUBSTITUTE

ANSWER

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MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

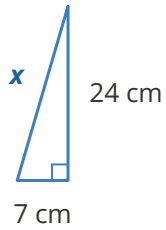
TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the hypotenuse, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Expand and simplify.

$$5(x + 6) + 2$$

METHOD Write the rule: **expanding and simplifying**

FROM MEMORY

WORKING

ANSWER

3 Calculate.

10% of \$120

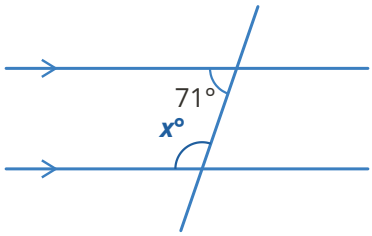
FORMULA Write the formula: **percentage of a quantity**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Write the formula: **co-interior angles**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Decrease 300 kg by 40%.

METHOD Write the rule: **percentage decrease**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5
1 Between which two whole numbers does this square root lie? $\sqrt{114}$	METHOD Write the rule: **estimating square roots** FROM MEMORY WORKING ANSWER	
2 The canteen buys a tray of 20 pies for $40 and sells it for $42. Find the profit as a percentage of the cost price.	FORMULA Write the formula: **profit and loss as a percentage** FROM MEMORY SUBSTITUTE ANSWER	
3 Simplify. Write your answer as a single power. $(7^4)^4$	FORMULA Write the formula: **power of a power** FROM MEMORY SUBSTITUTE ANSWER	
4 Does the point (2, 8) lie on the line $y = x + 6$? Explain. EXPLAIN	METHOD Write the rule: **points on a line** FROM MEMORY WORKING EXPLAIN	
5 Solve. $3x^2 = 108$	METHOD Write the rule: **equations of the form $ax^2 = c$** FROM MEMORY WORKING ANSWER	

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Factorise.

$$27y + 63$$

METHOD Write the rule: **factorising**

FROM MEMORY

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$2^3 \times 2^2$$

FORMULA Write the formula: **index law for multiplying**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Find the mode of these scores.

Scores	4	4	1	1	9	4	8	4

FORMULA Write the formula: **the mode**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Solve.

$$7n + 3 = 3n + 39$$

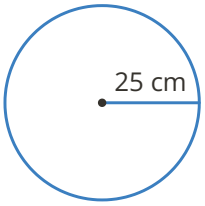
METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

5 Find the circumference of this circle, correct to 1 decimal place.



FORMULA Write the formula: **circumference of a circle (radius)**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Nina has saved $120, which is 50% of the cost of a new phone. How much does the phone cost?

METHOD Write the rule: **the unitary method**

FROM MEMORY

WORKING

ANSWER

2 Solve. Give your answer as a fraction or a decimal.

$$4a + 5 = 16$$

METHOD Write the rule: **solving equations**

FROM MEMORY

WORKING

ANSWER

3 Round to 2 decimal places.

$$5.6618$$

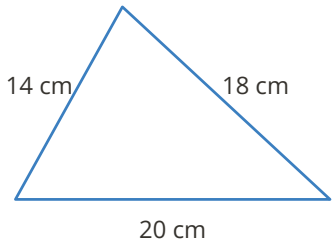
METHOD Write the rule: **rounding decimals**

FROM MEMORY

WORKING

ANSWER

4 Find the perimeter of this triangle.



METHOD Write the rule: **perimeter of a shape**

FROM MEMORY

WORKING

ANSWER

5 Evaluate.

$$7^2 + 6^0$$

FORMULA Write the formula: **the zero index**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$5(2y + 9)$$

FORMULA Write the formula: **expanding brackets**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Hiring a kayak at Cudgen Creek costs C dollars for h hours. Use the formula to find h when $C = 35$.

FORMULA $C = 5h + 20$

COPY

SUBSTITUTE

ANSWER

3 Simplify. Write your answer as a single power.

$$2^6 \div 2^3$$

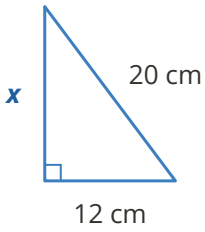
FORMULA Write the formula: **index law for dividing**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Heads came up 26 times in 40 tosses of a coin. Find the observed probability of heads.

FORMULA Write the formula: **observed (experimental) probability**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify.

$$4a^2 + 7a - 3a^2 + 2a$$

METHOD Write the rule: **like terms**

FROM MEMORY

WORKING

ANSWER

2 Priya thinks of a number, doubles it and adds 13. The answer is 21. Write and solve an equation to find the number.

METHOD Write the rule: **equation problems**

FROM MEMORY

WORKING

ANSWER

3 Sam says 6 + 2 × 12 = 96. Explain the mistake and give the correct answer. EXPLAIN

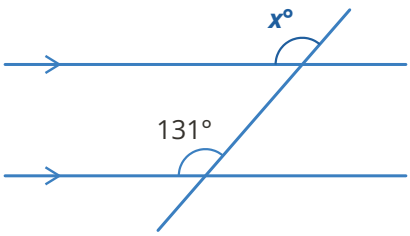
METHOD Write the rule: **order of operations**

FROM MEMORY

WORKING

EXPLAIN

4 Find the value of x.



FORMULA Write the formula: **corresponding angles**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Simplify. Write your answer as a single power.

$$(6^2)^4$$

FORMULA Write the formula: **power of a power**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Nina says $3^4 \times 3^4 = 9^8$. Explain the mistake. EXPLAIN

FORMULA Write the formula: **index law for multiplying**

FROM MEMORY

WORKING

EXPLAIN

2 Expand and simplify.

$$5(x + 2) + 2$$

METHOD Write the rule: **expanding and simplifying**

FROM MEMORY

WORKING

ANSWER

3 The probability that the bus is late is $\frac{3}{10}$.
Find the probability that the bus is not late.

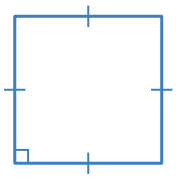
FORMULA Write the formula: **complementary events**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Find the area of this square.



5 cm

FORMULA Write the formula: **area of a square**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Increase $40 by 20%.

METHOD Write the rule: **percentage increase**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The price of a skateboard is $80 before GST. How much is the GST (10%)?

FORMULA Write the formula: **GST**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Solve.

$$8k + 11 = 6k + 29$$

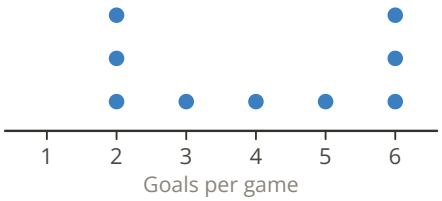
METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

3 The dot plot shows the goals a netball team scored in each game. Find the range.



FORMULA Write the formula: **the range**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Factorise.

$$7a + 14$$

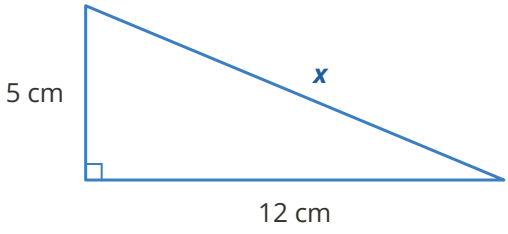
METHOD Write the rule: **factorising**

FROM MEMORY

WORKING

ANSWER

5 Find the length of the hypotenuse, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The price of a surf lesson changed from \$80 to \$88. Find the percentage increase.

FORMULA Write the formula: **percentage change**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Write 90 as a product of prime factors in index form.

METHOD Write the rule: **prime factors in index form**

FROM MEMORY

WORKING

ANSWER

3 Find the mean of these scores.

Scores	15	7	10	8

FORMULA Write the formula: **the mean**

FROM MEMORY

SUBSTITUTE

ANSWER

4 A triangle has sides 4 cm, 5 cm and 6 cm. Is it right-angled?

FORMULA Write the formula: **converse of Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Simplify.

$$\frac{15m}{3}$$

METHOD Write the rule: **dividing terms**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

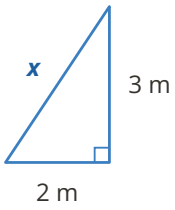
TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the hypotenuse, x, correct to 1 decimal place.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$6^7 \div 6^4$$

FORMULA Write the formula: **index law for dividing**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Solve.

$$x^2 = 49$$

METHOD Write the rule: **equations of the form $ax^2 = c$**

FROM MEMORY

WORKING

ANSWER

4 Find the range of these test marks.

Marks	59	72	83	72	79	58

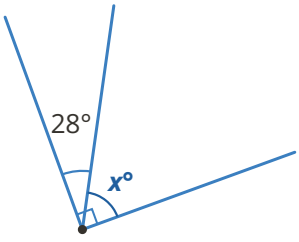
FORMULA Write the formula: **the range**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Write the formula: **complementary angles**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Expand.

$$8(5m + 4)$$

FORMULA Write the formula: **expanding brackets**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Decrease 150 kg by 50%.

METHOD Write the rule: **percentage decrease**

FROM MEMORY

WORKING

ANSWER

3 Which has the larger area: a 5 m by 4 m rectangle or a square with sides of 4 m? Explain. EXPLAIN

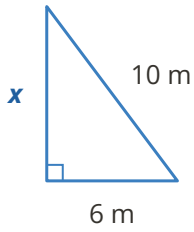
FORMULA Write the formula: **area of a rectangle and a square**

FROM MEMORY

WORKING

EXPLAIN

4 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Use the formula to find t when $v = 17$, $u = 5$ and $a = 2$.

FORMULA $v = u + at$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$6k + 4 = 2k + 28$$

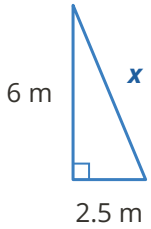
METHOD Write the rule: **pronumerals on both sides**

FROM MEMORY

WORKING

ANSWER

2 Find the length of the hypotenuse, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

3 The stem-and-leaf plot shows the ages of people at a beach. Find the range.

4	2 5
5	0 2 5 8
6	5 8

Key: 4 | 2 = 42

FORMULA Write the formula: **the range**

FROM MEMORY

SUBSTITUTE

ANSWER

4 A surf shop buys a cap for $20 and sells it for $30. Find the profit as a percentage of the cost price.

FORMULA Write the formula: **profit and loss as a percentage**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Zoe scored 6 out of 40 on a quiz. Write this as a percentage.

FORMULA Write the formula: **writing one amount as a percentage of another**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A school bag costs $90. It is on sale at 30% off. Find the sale price.

METHOD Write the rule: **discounts**

FROM MEMORY

WORKING

ANSWER

2 Solve. Give your answer as a fraction or a decimal.

$$2a + 9 = 22$$

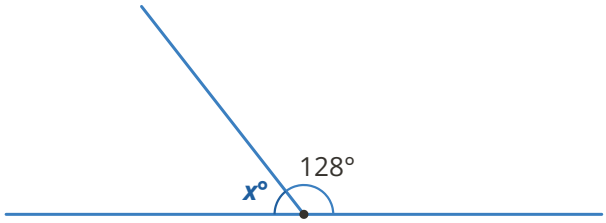
METHOD Write the rule: **solving equations**

FROM MEMORY

WORKING

ANSWER

3 Find the value of x.



FORMULA Write the formula: **angles on a straight line (supplementary)**

FROM MEMORY

SUBSTITUTE

ANSWER

4 Find the value of this expression when $a = -2$ and $b = -2$.

$$3a - b$$

METHOD Write the rule: **substitution**

FROM MEMORY

WORKING

ANSWER

5 A farm paddock covers 3 ha. Convert this to m^2.

FORMULA Write the formula: **area units**

FROM MEMORY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Theo thinks of a number, doubles it and adds 4. The answer is 26. Write and solve an equation to find the number.

METHOD Write the rule: **equation problems**

FROM MEMORY

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$(2^2)^3$$

FORMULA Write the formula: **power of a power**

FROM MEMORY

SUBSTITUTE

ANSWER

3 Calculate.

$$260 \times 10$$

METHOD Write the rule: **multiplying and dividing by 10, 100, 1000**

FROM MEMORY

WORKING

ANSWER

4 Factorise.

$$6x + 24$$

METHOD Write the rule: **factorising**

FROM MEMORY

WORKING

ANSWER

5 Between which two whole numbers does this square root lie?

$$\sqrt{87}$$

METHOD Write the rule: **estimating square roots**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

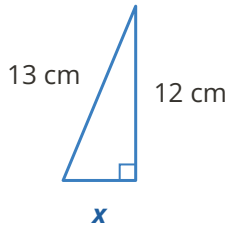
TERM 4 Recall each formula or rule, then use it.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the length of the shorter side, x.



FORMULA Write the formula: **Pythagoras' theorem**

FROM MEMORY

SUBSTITUTE

ANSWER

2 Zoe has saved \$64, which is 40% of the cost of a new phone. How much does the phone cost?

METHOD Write the rule: **the unitary method**

FROM MEMORY

WORKING

ANSWER

3 A bag holds 2 blue and 4 green marbles. One is chosen at random. Is it more likely than not to be blue? Explain. EXPLAIN

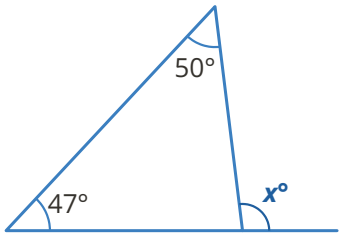
FORMULA Write the formula: **probability of an event**

FROM MEMORY

WORKING

EXPLAIN

4 Find the value of x.



FORMULA Write the formula: **exterior angle of a triangle**

FROM MEMORY

SUBSTITUTE

ANSWER

5 Simplify.

$$3p \times 2p$$

METHOD Write the rule: **multiplying and dividing terms**

FROM MEMORY

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Formula · Reference 1

TERM 4 Every formula and rule used in this booklet.

NUMBER

FRACTIONS AND PERCENTAGES OF A QUANTITY

Fraction of a quantity = fraction × quantity

Percentage of a quantity = percentage ÷ 100 × quantity

Percentage = part ÷ whole × 100%

ADDING, SUBTRACTING AND MULTIPLYING FRACTIONS

Same denominators: add or subtract the numerators, then simplify

INTEGERS

Subtracting a negative is the same as adding

MULTIPLYING AND DIVIDING BY 10, 100, 1000

× 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

ORDER OF OPERATIONS

Brackets, then × and ÷ (left to right), then + and − (left to right)

POWERS AND ROOTS

$a^n = a \times a \times \ldots \times a$ (n factors)

PRIME FACTORS (FACTOR TREE)

Use a factor tree: keep splitting until every branch ends in a prime

ROUNDING DECIMALS

Look at the next digit: 5 or more, round up; 4 or less, leave it

INDEX LAWS · YEAR 8

$a^m \times a^n = a^{m+n}$

$a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$

$a^0 = 1$ (any non-zero number to the power 0 is 1)

PERCENTAGE INCREASE AND DECREASE · YEAR 8

Percentage change = change ÷ original amount × 100%

Discount = % × price; sale price = price − discount

Find the percentage of the amount, then take it away

Find the percentage of the amount, then add it on

PROFIT, LOSS AND GST · YEAR 8

GST = 10% of the price; price incl. GST = price + GST

Profit (or loss) % = profit (or loss) ÷ cost price × 100%

PRIME FACTORS IN INDEX FORM · YEAR 8

Factor tree to primes, then write repeated primes as powers

THE UNITARY METHOD · YEAR 8

Unitary method: find 1% (or 10%) first, then multiply to get 100%

ALGEBRA AND EQUATIONS

LINEAR RELATIONSHIPS

Substitute each x-value into the rule

Coordinates (x, y): across first, then up or down

A point lies on a line if its coordinates make the rule true

MULTIPLYING AND DIVIDING TERMS

Divide the numbers, then cancel common pronumerals ($a^2 \div a = a$)

Divide the numbers, then cancel pronumerals that are top and bottom

Multiply the numbers, then the pronumerals ($a \times a = a^2$)

LIKE TERMS

Like terms have exactly the same pronumerals and powers

Only like terms (the same pronumerals) can be added or subtracted

WRITING ALGEBRAIC EXPRESSIONS

Turn words into symbols: more than +, less than −, times ×

EQUATION PROBLEMS · YEAR 8

Let n be the number; write an equation, then solve it

EQUATIONS OF THE FORM AX² = C · YEAR 8

$x^2 = c$ has 2 solutions: $x = \sqrt{c}$ or $x = -\sqrt{c}$

EXPANDING AND SIMPLIFYING · YEAR 8

Expand the brackets, then collect like terms

EXPANDING BRACKETS · YEAR 8

$a(b + c) = ab + ac$

FACTORISING · YEAR 8

Take out the highest common factor (HCF); check by expanding

PRONUMERALS ON BOTH SIDES · YEAR 8

Move the pronumerals to one side first, then solve

SOLVING EQUATIONS · YEAR 8

Do the same to both sides (use the inverse operation)

SUBSTITUTION · YEAR 8

Replace each pronumeral with its value (use brackets)

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 2

TERM 4 Every formula and rule used in this booklet (continued).

MEASUREMENT

AREA OF RECTANGLES AND SQUARES

$A = lw$ and $A = s^2$

$A = s^2$

AREA OF A COMPOSITE SHAPE

Split the shape into rectangles (and triangles), then add their areas

AREA OF A PARALLELOGRAM

$A = bh$

AREA OF A TRIANGLE

$A = \frac{1}{2}bh$

CIRCUMFERENCE OF A CIRCLE

$C = 2\pi r$

PERIMETER

Perimeter = the sum of all the side lengths

UNITS OF AREA, VOLUME AND CAPACITY

1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

VOLUME OF PRISMS

$V = lwh$

PYTHAGORAS' THEOREM · YEAR 8

Find the square numbers just below and just above the number

$c^2 = a^2 + b^2$ (c is the hypotenuse)

Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

GEOMETRY: ANGLES

ANGLES AT A POINT AND ON A LINE

Vertically opposite angles are equal

Angles at a point add to 360°

Complementary angles add to 90°

Angles on a straight line add to 180°

TRIANGLES AND QUADRILATERALS

Exterior angle = sum of the 2 interior opposite angles

Angle sum of a triangle = 180°

Angle sum of a quadrilateral = 360°

Isosceles triangle: base angles are equal; angle sum = 180°

ANGLES ON PARALLEL LINES

Alternate angles on parallel lines are equal

Co-interior angles on parallel lines add to 180°

Corresponding angles on parallel lines are equal

STATISTICS AND PROBABILITY

MEAN, MEDIAN, MODE AND RANGE

Range = highest score − lowest score

Mean = sum of scores ÷ number of scores

Median = middle score when the scores are in order (2 middles: halfway)

Mode = the most common score

PROBABILITY

P(E) = number of favourable outcomes ÷ total number of outcomes

Observed probability = frequency ÷ number of trials

P(not E) = 1 − P(E)

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

My · Tracker

TERM 4 Write your score after each session.

NAME _____

DATE _____

WEEK	SESSION 1	SESSION 2	SESSION 3	WEEK TOTAL
Week 1	/ 5	/ 5	/ 5	/ 15
Week 2	/ 5	/ 5	/ 5	/ 15
Week 3	/ 5	/ 5	/ 5	/ 15
Week 4	/ 5	/ 5	/ 5	/ 15
Week 5	/ 5	/ 5	/ 5	/ 15
Week 6	/ 5	/ 5	/ 5	/ 15
Week 7	/ 5	/ 5	/ 5	/ 15
Week 8	/ 5	/ 5	/ 5	/ 15
Week 9	/ 5	/ 5	/ 5	/ 15
Week 10	/ 5	/ 5	/ 5	/ 15
TERM TOTAL				**/ 150**

OUTCOMES IN THIS BOOKLET · NSW MATHEMATICS K–10 (2022), STAGE 4

MA4-INT-C-01	compares, orders and calculates with integers to solve problems	NCM 7
MA4-FRC-C-01	represents and operates with fractions, decimals and percentages to solve problems	NCM 7, NCM 8
MA4-IND-C-01	operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws	NCM 7, NCM 8
MA4-ALG-C-01	generalises number properties to operate with algebraic expressions including expansion and factorisation	NCM 7, NCM 8
MA4-EQU-C-01	solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$	NCM 8
MA4-LIN-C-01	creates and displays number patterns and finds graphical solutions to problems involving linear relationships	NCM 7
MA4-LEN-C-01	applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems	NCM 7
MA4-ARE-C-01	applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems	NCM 7
MA4-VOL-C-01	applies knowledge of volume and capacity to solve problems involving right prisms and cylinders	NCM 7
MA4-PYT-C-01	applies Pythagoras' theorem to solve problems in various contexts	NCM 8
MA4-ANG-C-01	applies angle relationships to solve problems, including those related to transversals on sets of parallel lines	NCM 7
MA4-GEO-C-01	identifies and applies the properties of triangles and quadrilaterals to solve problems	NCM 7
MA4-DAT-C-02	analyses simple datasets using measures of centre, range and shape of the data	NCM 7
MA4-PRO-C-01	solves problems involving the probabilities of simple chance experiments	NCM 7
MAO-WM-01	Working mathematically: the weekly reasoning questions (marked EXPLAIN or REASON) ask students to explain, check or justify.	

Year 8 units in these warm-ups (each starts once its teaching ends in the 2027 Year 8 mixed ability program): Rational numbers, percentages & financial maths from T1 Wk 5; Index laws from T1 Wk 7; Pythagoras' theorem from T1 Wk 10; Algebra 2 from T2 Wk 4; Equations 2 from T2 Wk 9. All other questions revise Year 7 (NCM 7 Ch 1–11). Ratios, rates and time (NCM 7 Ch 12) are held back until Year 8 teaches them.